

Unit 3, Lesson 6: Multiplying Numbers Expressed in Scientific Notation

Benchmark

MA.B.NSO.1.5 Add, subtract, multiply, and divide numbers expressed in scientific notation with procedural fluency.

Learning Target

- ☐ I can multiply numbers expressed in scientific notation.

3.6.1 Warm-Up: Bell Work**Add.**

1. $(4.79 \times 10^7) + (1.725 \times 10^5)$

2. $(2.9 \times 10^4) + (2.5 \times 10^4)$

3. $(1.7 \times 10^{-2}) + (1.3 \times 10^{-2})$

4. $(9.9 \times 10^{-7}) + (7.48 \times 10^{-9})$

3.6.2 Guided Instruction: Multiplying Numbers Expressed in Scientific Notation

Find the product of each.

1. $(7.8 \times 10^{14})(2.5 \times 10^{16})$
2. $(1.2 \times 10^{-9})(3.2 \times 10^{-7})$
3. Celia’s work for multiplying $(2 \times 10^5)(4 \times 10^5)$ is shown. She did the problem first using scientific notation and then checked her answer using standard notation. She noticed the answers do not match.

Scientific Notation	Standard Notation
$(2 \times 10^5)(4 \times 10^5)$	$200,000 \times 400,000$
$(2 \times 4)(10^5 \times 10^5)$	$8 \times 10,000,000,000$
8×10^5	$80,000,000,000$

Find Celia’s mistake, then write her a note to help her learn from her mistake.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

3.6.3 Guided Practice: Multiplying Numbers Expressed in Scientific Notation

1. Determine the product of $(3.1 \times 10^{-9})(3.2 \times 10^{-7})$.
2. Check your work by finding the product of the numbers when written in standard form.
3. Complete the work to find the product. Be sure to write your answer in scientific notation.

$$\begin{aligned} & (8.8 \times 10^4)(4.92 \times 10^5) \\ &= (\quad) (10^4 \cdot 10^5) \\ &= 43.296 \times 10 \\ &= \underline{\hspace{2cm}} \times 10^{10} \end{aligned}$$

3.6.4 Activity: Give Me Five!

You and your partner will be given a set of problem cards and a gameboard. Your goal is to get five squares in a row, horizontally, vertically, or diagonally, marked on the gameboard. To mark a square, you need to correctly find the product of a given problem.

Record each problem you complete on the recording sheet. Not all of the spaces must be filled in order for a winner to be determined.

Give Me Five! - Recording Sheet

Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:
Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:

Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:
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Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:	Problem #: ____ Answer: ____ Work:

3.6.5 Wrap-Up: Growth Mindset

Which mindset did I demonstrate?	Choose the best description.
Did I use the partner discussions as learning opportunities?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I work on problems that challenged me?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I use strategies to 'un-stick' myself when I found the problems difficult?	☐ Never ☐ Sometimes ☐ Usually ☐ Always
Did I put as much effort as I possibly could into the problems?	☐ Never ☐ Sometimes ☐ Usually ☐ Always

Materials Needed:

2 dice

Set of problem cards

Game Play:

To determine who goes first, each partner rolls a die, and the partner with the lower number goes first.

Roll the dice and use the digits to make a 2-digit number. For example, if a '2' and a '3' are rolled, the number made could be either '23' or '32'. If either number rolled is a '6,' the player loses a turn.

Select the card with the number you created in step 2 and solve the problem, recording your work and the answer on the recording sheet in the workbook. Then mark the corresponding square on the gameboard.

Play continues until one partner has five squares in a row, column, or diagonal marked off on the gameboard. Both partners check using the answer sheet or a calculator. If all problems are correct, the player wins.

Problem Cards

11 $(9.6 \times 10^{12})(8.2 \times 10^{12})$	21 $(6.44 \times 10^6)(2.3 \times 10^5)$	31 $(9.45 \times 10^9)(3.2 \times 10^9)$
12 $(9.2 \times 10^{-12})(3.2 \times 10^{-10})$	22 $(1.1 \times 10^{-19})(3.6 \times 10^{-19})$	32 $(2.47 \times 10^{-8})(1.9 \times 10^{-7})$
13 $(1.32 \times 10^{10})(3.4 \times 10^9)$	23 $(4.04 \times 10^{-6})(2.6 \times 10^{-4})$	33 $(8.25 \times 10^{-1})(3.3 \times 10^1)$
14 $(2.07 \times 10^8)(9 \times 10^7)$	12 $(5.46 \times 10^8)(4 \times 10^8)$	34 $(6.96 \times 10^7)(2.4 \times 10^5)$
15 $(1.1 \times 10^{-10})(2.9 \times 10^{-10})$	25 $(1.521 \times 10^{22})(3.9 \times 10^{20})$	35 $(7.3 \times 10^{10})(1.25 \times 10^{10})$

41 $(5.3 \times 10^{14})(3.8 \times 10^{12})$	51 $(5.04 \times 10^{-3})(1.1 \times 10^{-2})$	53 $(3.25 \times 10^{-5})(1.3 \times 10^{-5})$
42 $(4.72 \times 10^5)(1.18 \times 10^5)$	52 $(1.55 \times 10^{12})(3.1 \times 10^{11})$	54 $(1.24 \times 10^8)(3.8 \times 10^7)$
43 $(2.16 \times 10^{-3})(1.2 \times 10^{-9})$	45 $(3.6 \times 10^{-11})(2.43 \times 10^{-10})$	55 $(3.816 \times 10^{-10})(1.59 \times 10^{-8})$
44 $(1.01 \times 10^{-6})(2.6 \times 10^{-4})$		

Gameboard

11	21	31	41	51
12	22	32	42	52
13	23	33	43	53
14	24	34	44	54
15	25	35	45	55

Give Me Five! – Answer Sheet

11	(7.872×10^{27})	21	(1.4812×10^{12})	31	(3.024×10^{19})
12	(2.944×10^{-21})	22	(3.96×10^{-36})	32	(4.693×10^{-15})
13	(4.488×10^{18})	23	(1.0504×10^{-9})	33	(2.7225×10^1)
14	(1.863×10^{16})	24	(2.184×10^{17})	34	(1.6704×10^{12})
15	(3.19×10^{-20})	25	(5.9319×10^{42})	35	(9.125×10^{20})

41	(2.014×10^{27})	51	(5.544×10^{-5})
42	(5.5696×10^{10})	52	(4.805×10^{23})
43	(2.592×10^{-18})	53	(4.225×10^{-10})
44	(2.626×10^{-10})	54	(4.712×10^{15})
45	(8.748×10^{-21})	55	$(6.06744 \times 10^{-18})$